Rubric and Mark Scheme: Flood-Warning Sensor Model

Total: 20 marks | Award method marks where a correct procedure is shown

General marking principles

Accept equivalent notation and algebra. Carry forward a student's earlier numerical value when the later method is correct, unless the earlier value makes the later result impossible. Do not double-penalize the same conceptual error. Award explanation marks only when the meaning is stated in context.

1. Alarm probability and diagnosis - 7 marks

1(a) [3] Recognizes the partition H and N and writes $P(A)=P(A|H)P(H)+P(A|N)P(N)$. [1] Substitutes $0.92(0.08)+0.06(0.92)$. [1] Obtains 0.1288. [1]

1(b) [4] Uses Bayes rule $P(H|A)=P(A|H)P(H)/P(A)$. [1] Correct numerator $0.92(0.08)=0.0736$. [1] Calculates $0.0736/0.1288=0.5714$ (about 57.1%). [1] Explains that sensitivity conditions on high water, whereas $P(H|A)$ also depends on the prior/base rate and false alarms. [1]

Common error: reporting 0.92 for $P(H|A)$. Maximum 1/4 for part (b) if no Bayes reversal is attempted.

2. Discrete random variable - 5 marks

2(a) [2] States $X \sim \text{Binomial}(n=4, p=0.1288)$. One mark for the family and one for both correct parameters.

2(b) [2] Uses $C(4,1)(0.1288)(0.8712)^3$. [1] Obtains approximately 0.340. [1]

2(c) [1] $E[X]=np=4(0.1288)=0.5152$.

Allow follow-through from the student's value of $P(A)$. If the student uses $p=0.08$, award method marks but not the parameter accuracy mark.

3. Continuous measurement error - 5 marks

3(a) [2] Gives $f_E(e)=1/4$ for $-2<=e<=2$. [1] Gives $f_E(e)=0$ otherwise or clearly states the support. [1]

3(b) [2] Recognizes the interval length from -0.5 to 0.5 is 1. [1] Probability = area = 1 x 1/4 = 0.25. [1]

3(c) [1] Median = 0, with or without symmetry justification.

4. Modelling judgement - 3 marks

Full-credit response. Identifies a plausible dependence mechanism, such as persistent weather conditions, river inertia, serial correlation, or an alarm state carrying over to the next hour. [1] Explains that neighboring hours then have related probabilities rather than independent trials. [1] States a consequence, such as underestimated variability, misleading tail probabilities, too many/few predicted alarm clusters, or overconfident operational planning. [1]

Examples of acceptable answers

- During a storm, high water tends to persist. Alarms in adjacent hours are positively correlated, so a binomial model can underestimate the chance of long runs of alarms.
- Sensor malfunction can persist across hours. Treating each hour independently may underestimate the probability of repeated false alarms.
- River level trends over time, so the marginal alarm probability is not constant. Using one fixed p can bias expected alarm counts.

Holistic performance descriptors

Band	Marks	Description
Excellent	17-20	Accurate calculations, explicit conditioning, correct models and supports, and a context-aware critique of assumptions.
Good	13-16	Mostly correct methods with minor arithmetic or notation slips; interpretation is generally sound.
Developing	8-12	Some correct procedures, but confusion about reversed conditioning, model parameters, or continuous probability remains.
Insufficient	0-7	Fragmentary work, unsupported answers, or major misconceptions across several sections.

Feedback prompts for markers

Use comments that identify the precise next step: “You calculated P(A|H), but the question asks for P(H|A); write Bayes rule before substituting,” or “Your density height is correct; now include the support and zero elsewhere.” Avoid comments that merely say “wrong formula.”

Answer summary

P(A)=0.1288; P(H|A)=0.5714; X~Binomial(4,0.1288); P(X=1) approximately 0.340; E[X]=0.5152; $f_E(e)$=1/4 on [-2,2]; P(|E|<=0.5)=0.25; median(E)=0.